



Supporting Information

Why Mg_2IrH_6 Is Predicted to Be a High-Temperature Superconductor, But Ca_2IrH_6 Is Not

*X. Wang, W. E. Pickett, M. Hutcheon, R. P. Prasankumar, E. Zurek**

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Xiaoyu Wang, Warren E. Pickett, Micheal Hutcheon, Rohit Prasankumar, and Eva Zurek*

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S1 Computational Details

S1.1 Periodic Geometry Optimizations

Geometry optimizations and electronic structure calculations were performed via density functional theory (DFT) using the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional¹ as implemented in the Vienna *Ab Initio* Simulation Package (VASP).² The magnesium $2p^63s^2$, calcium $3p^64s^2$, hydrogen $1s^1$, and iridium $5d^76s^2$ electrons were treated explicitly via a plane-wave basis set with a kinetic energy cutoff of 900 eV, while the core states were described with projector augmented wave (PAW) potentials.³ The reciprocal space was sampled over a $13 \times 13 \times 13$ Γ -centered k -mesh for geometry optimizations, and a $18 \times 18 \times 18$ k -mesh for the density of states calculations.

S1.2 Electron Phonon Calculations

The phonon dynamical matrix and the electron-phonon coupling (EPC) parameters were obtained via the Quantum Espresso (QE) suite.⁴ The PSLibrary⁵ PAW potentials were used with valence configurations of $2s^22p^63s^2$ for magnesium, $3s^23p^64s^2$ for calcium, $5d^76s^2$ for iridium, and $1s^1$ for hydrogen. Following a previous study,⁶ the kinetic energy cutoff of the plane waves was set to 80 Ry. A $6 \times 6 \times 6$ q -mesh and a $18 \times 18 \times 18$ k -mesh were employed for phonon calculations, and were further interpolated onto a finer $36 \times 36 \times 36$ grid for the EPC.

The superconducting critical temperature (T_c) was estimated using the Allen-Dynes modified McMillan equation:⁷

$$T_c = \frac{f_1 f_2 \omega_{\text{ln}}}{1.20} \exp \left[-\frac{1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right], \quad (1)$$

in which the Coulomb renormalization parameter, μ^* , was set to 0.1, the logarithmic average phonon frequency ω_{ln} was obtained by

$$\omega_{\text{ln}} = \exp \left(\frac{2}{\lambda} \int \frac{d\omega}{\omega} \alpha^2 F(\omega) \ln \omega \right), \quad (2)$$

the electron phonon coupling constant, λ , was evaluated by

$$\lambda = \int d\omega \alpha^2 F(\omega) / \omega, \quad (3)$$

and the shape correction parameters as well as the EPC parameters are presented in Table S1. The Eliashberg spectral function, $\alpha^2 F(\omega)$, was obtained from the QE calculations, and was also used to numerically solve the isotropic Eliashberg equations.⁸

Structure	λ	ω_{ln} (K)	f_1	f_2	T_c (K)
Mg ₂ IrH ₆ (ambient)	2.06	436	1.14	1.11	80.5 (102.2)
Ca ₂ IrH ₆ (ambient)	0.25	435	1.01	1.00	0.02
Ca ₂ IrH ₆ (25 GPa)	0.24	537	1.01	1.00	0.02

Table S1: The electron phonon coupling (λ), logarithmic average frequency (ω_{ln}), strong coupling (f_1) and shape dependence correction factors (f_2), and the superconducting critical temperature (T_c), estimated using the Allen-Dynes modified McMillan equation with $\mu^* = 0.1$. For Mg₂IrH₆ the numerical solution of the Eliashberg equations was employed to obtain the value in brackets.

The phonon linewidth for a certain phonon branch ν at a given wave vector $\mathbf{q}$ is obtained via

$$\gamma_{\mathbf{q}\nu} = 2\pi\omega_{\mathbf{q}\nu} \sum_{\mathbf{k}} |g_{\mathbf{k}+\mathbf{q},\mathbf{k}}^{\mathbf{q}\nu}|^2 \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\text{F}}) \delta(\varepsilon_{\mathbf{k}} - \varepsilon_{\text{F}}), \quad (4)$$

in which $\omega_{\mathbf{q}\nu}$ is the eigenvalue of the dynamical matrix and $g_{\mathbf{k}+\mathbf{q},\mathbf{k}}^{\mathbf{q}\nu}$ is the EPC matrix element obtained from the QE calculations. The Eliashberg spectral function can be obtained via the summation of $\gamma_{\mathbf{q}\nu}$ over the Brillouin zone by

$$\alpha^2 F(\omega) = \frac{1}{2\pi g_{\text{F}}} \sum_{\mathbf{q}\nu} \frac{\gamma_{\mathbf{q}\nu}}{\omega_{\mathbf{q}\nu}} \delta(\omega - \omega_{\mathbf{q}\nu}), \quad (5)$$

where g_{F} is the density of states at the Fermi level. It's possible to estimate the contribution towards the T_c from each vibrational mode by making the width of the line of the phonon dispersion curve at respected $\mathbf{q}$ and ν proportional to $\gamma_{\mathbf{q}\nu}/\omega_{\mathbf{q}\nu}$. This is the methodology used to determine the width of the line in Figure 3 of the main text, and Figure S4 in the SI.

S1.3 Molecular Calculations

Molecular calculations were performed via the Amsterdam Density Functional (ADF) package⁹ using the PBE functional¹ and a Slater-type triple- ζ polarized (TZP) basis set.¹⁰ A restricted open-shell electronic configuration was applied for the IrH₆⁴⁻ molecule. The resulting molecular orbitals are plotted in Figure 1a in the main text.

S2 Density of States with a Hybrid Functional

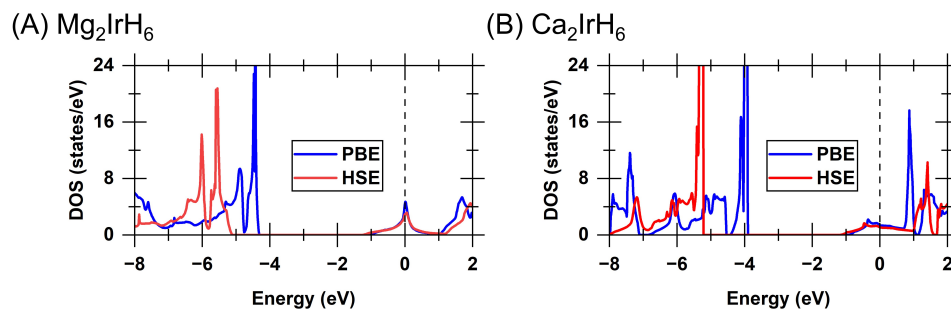


Figure S1: Comparison of the density of states (DOS) calculated using PBE and the range-separated hybrid HSE06 functional¹¹ for Mg_2IrH_6 and Ca_2IrH_6 at ambient pressure.

S3 Projected Bandstructures

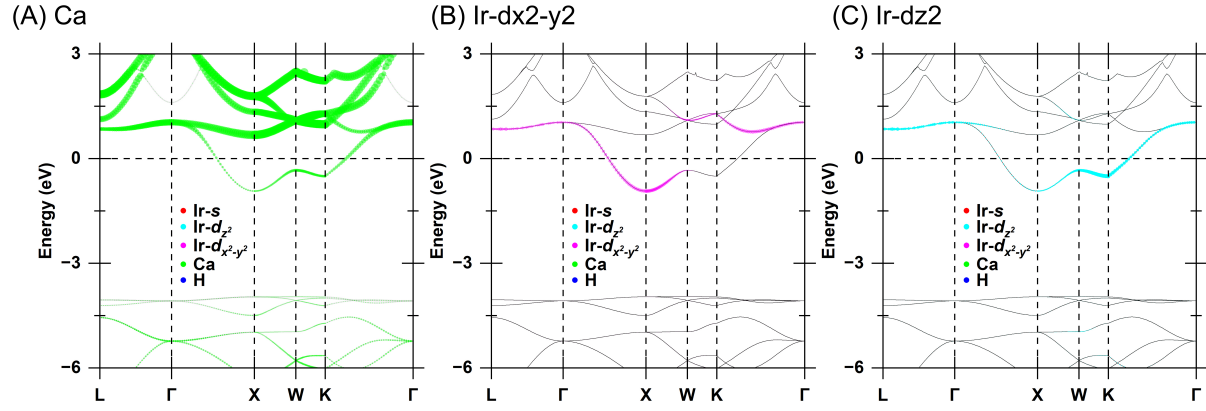


Figure S2: Projected bandstructure of Ca_2IrH_6 . Radius of the circles are proportional to the projections with the same scale in all plots.

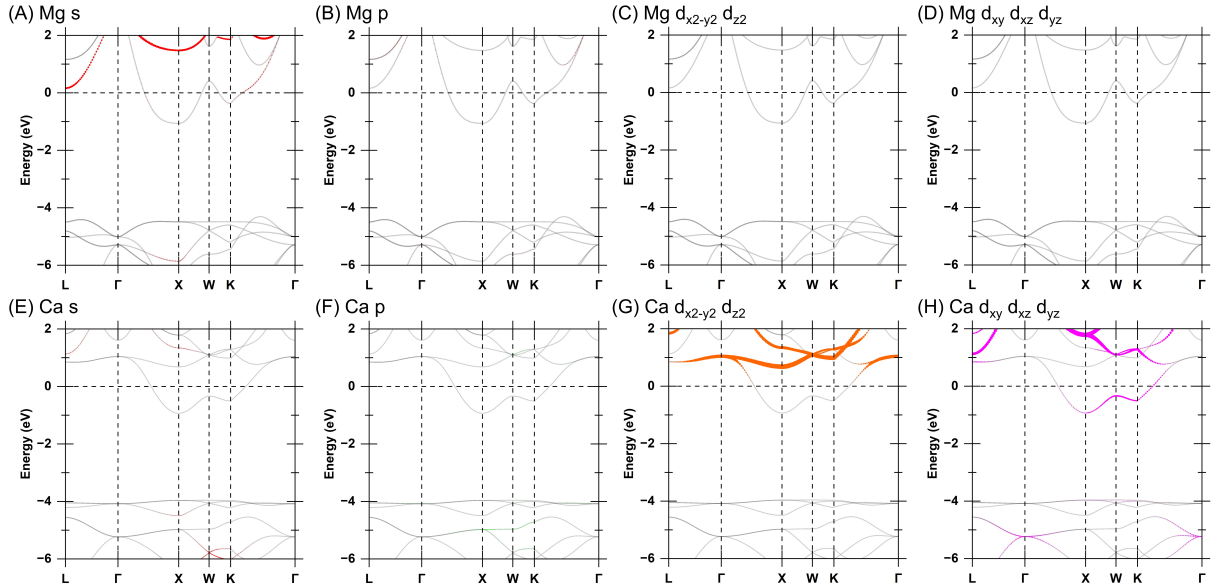


Figure S3: Electronic bandstructure of (A-D) Mg_2IrH_6 projected onto Mg and (E-H) Ca_2IrH_6 projected onto Ca orbitals. Radius of the circles are proportional to the projections with the same scale in all plots.

S4 Phonon Dispersion Curves, Density of States, and Electron-Phonon Coupling Properties

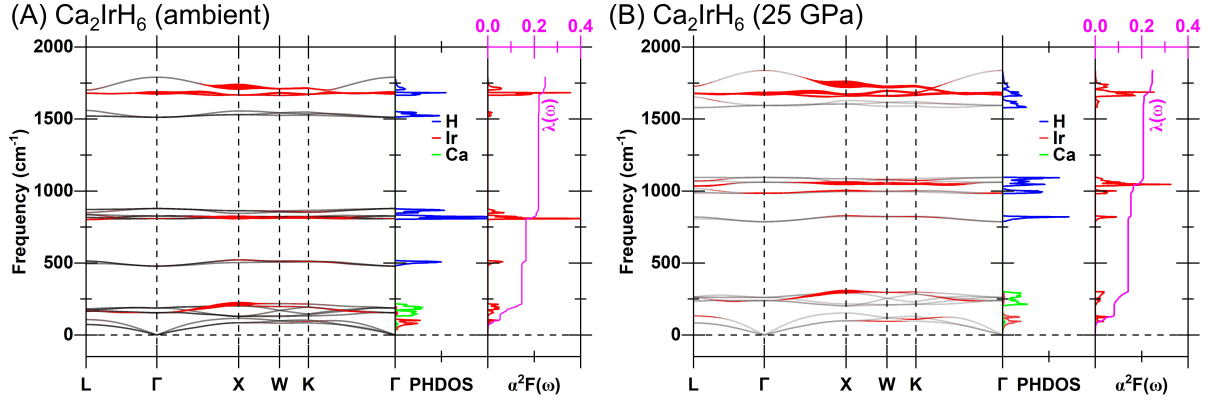


Figure S4: Phonon bandstructure, projected phonon density of states, Eliashberg spectral function, $\alpha^2F(\omega)$, and integral of the electron-phonon-coupling, $\lambda(\omega)$. The thickness of the line in the phonon bandstructure is proportional to $\lambda_{q\nu} \times \omega_{q\nu}$. (A) Ca_2IrH_6 at 0 GPa and (B) Ca_2IrH_6 at 25 GPa

S5 Fermi Surface

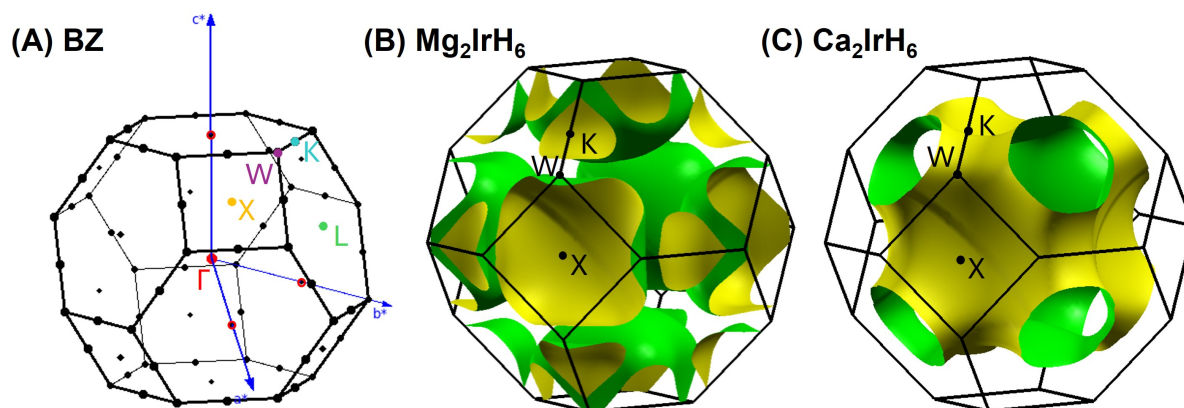


Figure S5: (A) First Brillouin zone of Mg_2IrH_6 and Ca_2IrH_6 , (B) Fermi surface of Mg_2IrH_6 , and (C) Fermi surface of Ca_2IrH_6 . High symmetry points are related to the labels in the bandstructure plots.

S6 Electron-Phonon Coupling in Ca_2IrH_6

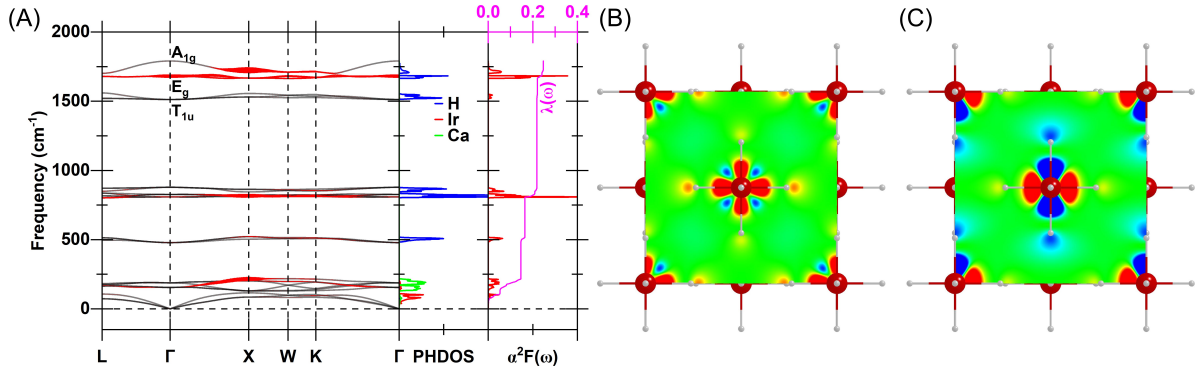


Figure S6: (A) Phonon bandstructure, projected phonon density of states, Eliashberg spectral function, $\alpha^2F(\omega)$, and integral of the electron-phonon-coupling, $\lambda(\omega)$, for Ca_2IrH_6 . The thickness of the line at any given wavevector $\mathbf{q}$ and phonon branch index ν is proportional to the ratio of the phonon linewidth to the phonon frequency ($\gamma_{\mathbf{q}\nu}/\omega_{\mathbf{q}\nu}$) (as described in Section S1.2). (B) Difference in the partial charge density for the occupied part of the metallic band between E_g -phonon-modulated structure and the unperturbed structure in the view of the xy plane and (C) yz plane. Charge gain/loss are represented with red/blue colors, while unchanged area is in green. The charge density difference plots resemble those computed for Mg_2IrH_6 (Figure 3, main text) around the IrH_6 unit, though the red and blue regions around hydrogen are significantly smaller, in-line with the lack of hydrogen character observed in the charge density of the occupied metal band (Figure 2C, main text). Further, there is very little change in the interstitial space, due to the depletion of the interstitial charge density by the backdonation to the Ca d -orbitals.

S7 Superconducting Critical Temperatures of Systems With and Without d -Backdonation

Table S2: Comparison of the predicted superconducting critical temperatures, T_c s, of superconductors reported in the literature. We provide data for systems with the same structural prototype listing both those that are unlikely ($M = \text{Mg}$), and likely ($M = \text{Ca}, \text{Sc}, \text{Y}$) to undergo d -backdonation. The T_c values reported were calculated via the Allen-Dynes McMillan equation. Values in parentheses were obtained via the numerical solution of the Eliashberg equations.

General Formula	Metal 1 ("M")	Metal 2 ("X")	Pressure (GPa)	T_c (K)
MH ₆	Mg		300	263, ¹² 260 (420) ¹³
MH ₆	Ca		250	187 ¹⁴
MH ₆	Sc		300	135 (169) ¹⁵
MH ₆	Y		120	251 ¹⁶
<i>Cmcm</i> -MH ₄	Mg		100	37 ¹⁷
<i>Cmcm</i> -MH ₄	Ca		200	12 ¹⁸
<i>Fd3m</i> -Li ₂ MH ₁₆	Mg		250	(473) ¹⁹
<i>Fd3m</i> -Li ₂ MH ₁₆	Ca		350	(330) ²⁰
M ₂ XH ₆	Mg	Ir	ambient	65, ²¹ (77), ²² (160) ⁶
M ₂ XH ₆	Ca	Ir	ambient	0.024, 0 ²²
M ₂ XH ₆	Mg	Rh	ambient	(48.5) ²²
M ₂ XH ₆	Ca	Rh	ambient	0 ²²
M ₂ XH ₆	Mg	Pd	ambient	(66.5) ²²
M ₂ XH ₆	Mg	Pt	ambient	(80.4) ²²
M ₂ XH ₆	Ca	Pd	ambient	(≤ 15) ²²
M ₂ XH ₆	Ca	Pt	ambient	(≤ 20) ²²
M ₂ C ₂ H ₈	Mg		50	35 ²³
M ₂ C ₂ H ₈	Ca		50	pseudogap ²³

S8 Structural Coordinates

Element	Multiplicity	Wyckoff Label	Fractional Coordinates		
Mg	8	c	0.250	0.250	0.250
Ir	4	a	0.000	0.000	0.000
H	24	e	0.741	0.000	0.000

Table S3: Structural coordinates of Mg_2IrH_6 at ambient pressure. Spacegroup $Fm\bar{3}m$ (225), $a = 6.661 \text{ \AA}$.

Element	Multiplicity	Wyckoff Label	Fractional Coordinates		
Ca	8	c	0.250	0.250	0.250
Ir	4	a	0.000	0.000	0.000
H	24	e	0.237	0.000	0.000

Table S4: Structural coordinates of Ca_2IrH_6 at ambient pressure. Spacegroup $Fm\bar{3}m$ (225), $a = 7.260 \text{ \AA}$.

Element	Multiplicity	Wyckoff Label	Fractional Coordinates		
Ca	8	c	0.250	0.250	0.250
Ir	4	a	0.000	0.000	0.000
H	24	e	0.257	0.000	0.000

Table S5: Structural coordinates of Mg_2IrH_6 at 25 GPa. Spacegroup $Fm\bar{3}m$ (225), $a = 6.648 \text{ \AA}$.

Element	Cartesian Coordinates		
Ir	0.000	0.000	0.000
H	1.741	0.000	0.000
H	-1.741	0.000	0.000
H	0.000	1.741	0.000
H	0.000	-1.741	0.000
H	0.000	0.000	1.741
H	0.000	0.000	-1.741

Table S6: Structural coordinates of the IrH_6^{3-} molecule.

Element	Cartesian Coordinates		
Ir	0.000	0.000	0.000
H	1.805	0.000	0.000
H	-1.805	0.000	0.000
H	0.000	1.805	0.000
H	0.000	-1.805	0.000
H	0.000	0.000	1.805
H	0.000	0.000	-1.805

Table S7: Structural coordinates of the IrH_6^{4-} molecule.

References

- [1] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, 77, 3865.
- [2] G. Kresse, J. Hafner, *Phys. Rev. B.* **1993**, 47, 558.
- [3] P. E. Blochl, *Phys. Rev. B* **1994**, 50, 17953.
- [4] P. Giannozzi, O. Andreussi, T. Brumme, O. Bunau, M. B. Nardelli, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, M. Cococcioni, *J. Phys. Cond. Mat.* **2017**, 29, 465901.
- [5] A. Dal Corso, *Comput. Mater. Sci.* **2014**, 95, 337–350.
- [6] K. Dolui, L. J. Conway, C. Heil, T. A. Strobel, R. P. Prasankumar, C. J. Pickard, *Phys. Rev. Lett.* **2024**, 132, 166001.
- [7] P. B. Allen, R. C. Dynes, *Phys. Rev. B* **1975**, 12, 905.
- [8] G. M. Eliashberg, *Sov. Phys. JETP* **1960**, 11, 696–702.
- [9] G. T. Te Velde, F. M. Bickelhaupt, E. J. Baerends, C. Fonseca Guerra, S. J. A. van Gisbergen, J. G. Snijders, T. Ziegler, *J. Comput. Chem.* **2001**, 22, 931–967.
- [10] E. Van Lenthe, E. J. Baerends, *J. Comput. Chem.* **2003**, 24, 1142–1156.
- [11] A. V. Krukau, O. A. Vydrov, A. F. Izmaylov, G. E. Scuseria, *J. Chem. Phys.* **2006**, 125, 224106.
- [12] X. Feng, J. Zhang, G. Gao, H. Liu, H. Wang, *RSC Adv.* **2015**, 5, 59292–59296.
- [13] R. Szczesniak, A. Durajski, *Front. Phys.* **2016**, 11, 1–5.
- [14] H. Wang, J. S. Tse, K. Tanaka, T. Iitaka, Y. Ma, *Proc. Natl. Acad. Sci.* **2012**, 109, 6463–6466.
- [15] X. Ye, N. Zarifi, E. Zurek, R. Hoffmann, N. Ashcroft, *J. Phys. Chem. C* **2018**, 122, 6298–6309.
- [16] Y. Li, J. Hao, H. Liu, J. S. Tse, Y. Wang, Y. Ma, *Sci. Rep.* **2015**, 5, 9948.
- [17] D. C. Lonie, J. Hooper, B. Altintas, E. Zurek, *Phys. Rev. B* **2013**, 87, 054107.
- [18] Z. Shao, D. Duan, Y. Ma, H. Yu, H. Song, H. Xie, D. Li, F. Tian, B. Liu, T. Cui, *Inorg. Chem.* **2019**, 58, 2558–2564.
- [19] Y. Sun, J. Lv, Y. Xie, H. Liu, Y. Ma, *Phys. Rev. Lett.* **2019**, 123, 097001.
- [20] M. Redington, E. Zurek, *Chem. Mater.* **2024**, 36, 8412–8423.
- [21] F. Zheng, Z. Zhang, Z. Wu, S. Wu, Q. Lin, R. Wang, Y. Fang, C.-Z. Wang, V. Antropov, Y. Sun *et al.*, *Mater. Today Phys.* **2024**, 42, 101374.

- [22] A. Sanna, T. F. Cerqueira, Y.-W. Fang, I. Errea, A. Ludwig, M. A. Marques, *npj Comput. Mater.* **2024**, *10*, 44.
- [23] N. Geng, K. P. Hilleke, F. Belli, P. K. Das, E. Zurek, *Mater. Today Phys.* **2024**, *44*, 101443.